

Begin Your Journey from Concept to Creation

TIPS FOR SUCCESS



LAB PROTOCOL

Sensors & Actuators Laboratory

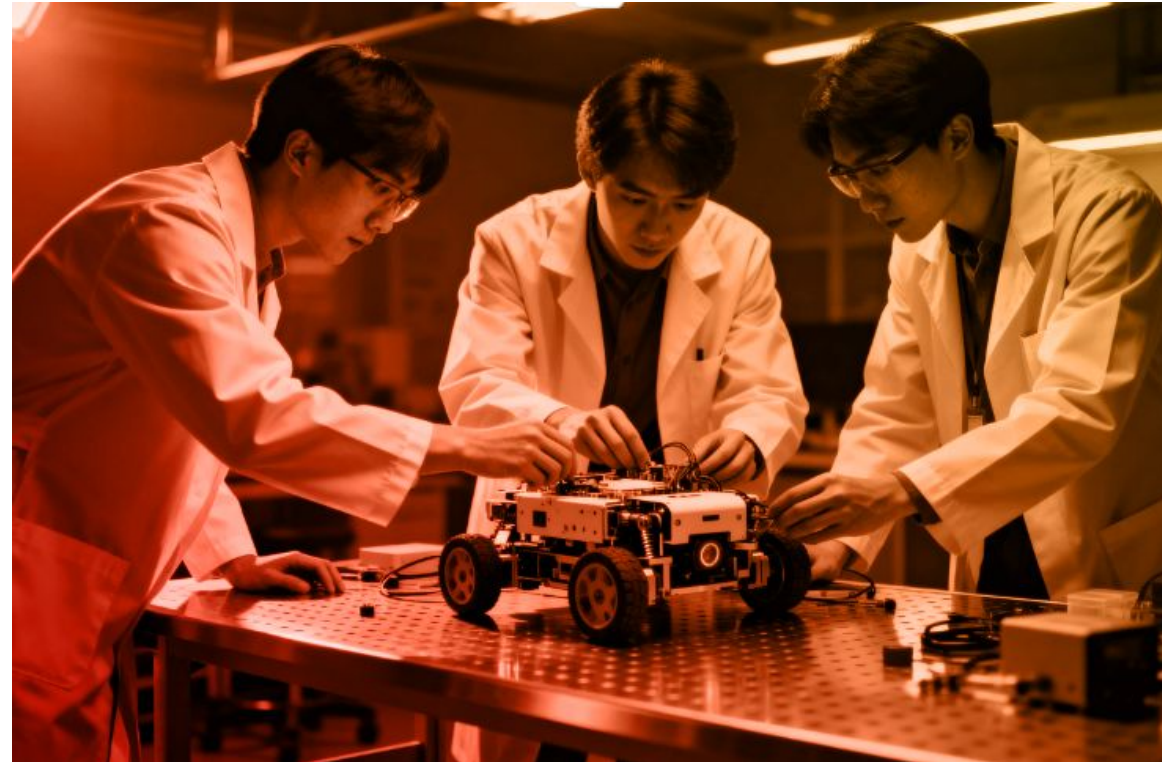
Theory into Practice



“Robotics is not just about building robots,
it's about building the *future*.
Start today. *The future is yours!*”



Course Code: 25PC2RA202
**Number of Lecture Hours per
week:** 2
Number of Credits: 1



**:RESOURCE
PERSONS:**

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STEP-BY-STEP ROADMAP TO LEARN ROBOTICS IN 2026

— A clear path for beginners to become future-ready in robotics! —



2026

YOUR YEAR TO
BUILD THE FUTURE

01

BUILD A STRONG FOUNDATION



- Learn basic electronics
- Understand circuits, voltage, current, resistance
- Learn Digital Logic & Microcontrollers

🕒 Time: 2-3 Weeks

02

LEARN PROGRAMMING



- Start with C/C++
- Learn programming logic
- Practice problem solving
- Platforms: Arduino IDE & VS Code

🕒 Time: 3-4 Weeks

03

WORK WITH ARDUINO



- Learn Arduino UNO
- Read sensors & control actuators
- Blink LED to building interactive projects

🕒 Time: 4-6 Weeks

04

LEARN SENSORS & ACTUATORS



- Work with IR, Ultrasonic, Gyro, Line Follower, etc.
- Understand Motors, Servos, Relays
- Interface different modules

🕒 Time: 4 Weeks

05

BUILD ROBOTICS PROJECTS



- Start with simple robots
- Obstacle Avoider
- Line Follower
- Bluetooth Controlled Car
- Robotic Arm

🕒 Time: 2-3 Months

06

LEARN ROBOTICS CONCEPTS



- Kinematics & Dynamics (Basics)
- Degrees of Freedom (DOF)
- Robot Mechanics
- Control Systems (Basics)

🕒 Time: 4-6 Weeks

07

EXPLORE ADVANCED TOPICS



- PID Control
- Computer Vision (OpenCV)
- SLAM Basics
- ROS (Robot Operating System) Introduction

🕒 Time: 2-3 Months

08

WORK ON REAL-WORLD PROJECTS



- Apply your skills
- Build problem-solving projects
- Participate in competitions & hackathons

🕒 Time: Ongoing

09

KEEP LEARNING & BUILD YOUR BRAND



- Follow robotics communities
- Read, Watch, Practice
- Share your projects
- Start your GitHub/Portfolio

🕒 Time: Lifetime



YOU'RE
FUTURE-READY!

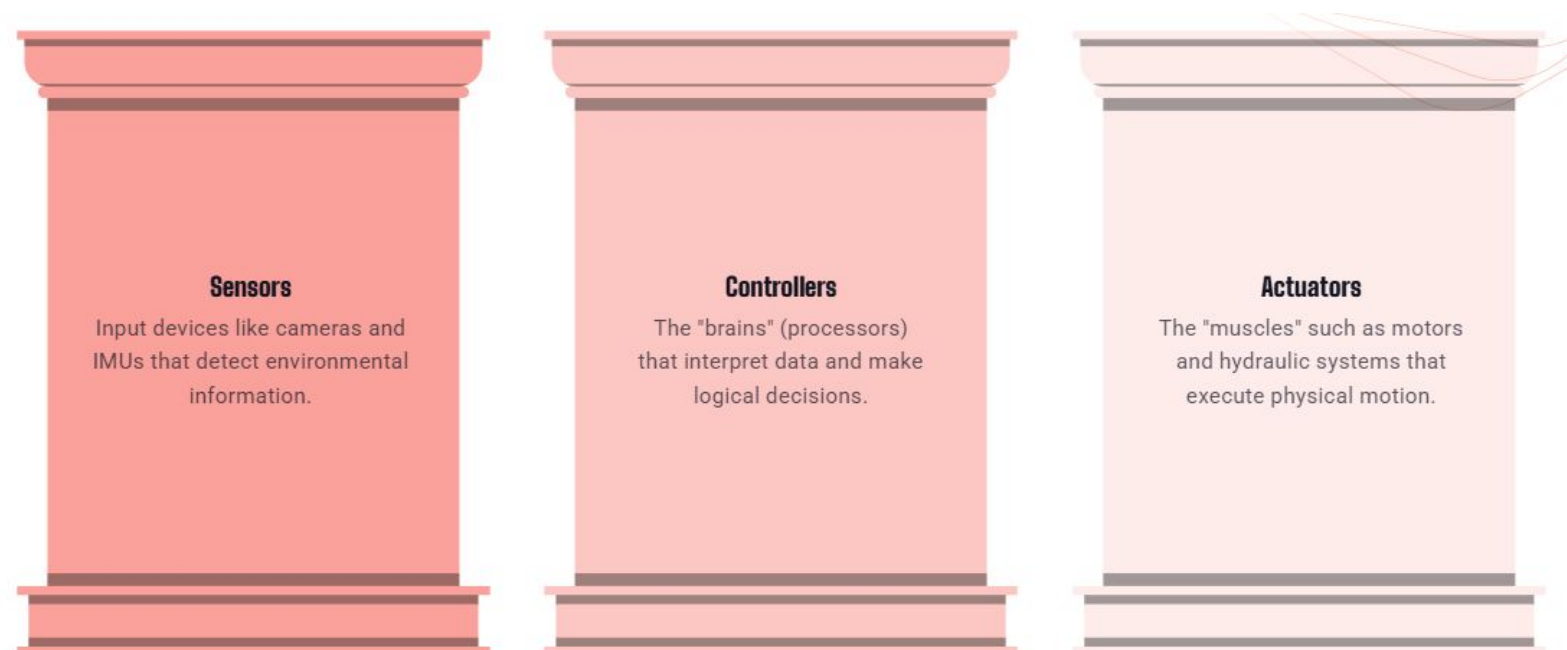
KEEP INNOVATING.
KEEP BUILDING.
KEEP GROWING!

- A **robot** is a machine that can perform tasks automatically by sensing its environment, processing information, and taking actions. Robots are typically programmable, allowing them to carry out different tasks with little or no human intervention.

Most robots have three main components:

- **Sensors** to detect information (such as cameras, microphones, or touch sensors),
- **A controller** (a computer or processor) to make decisions,
- **Actuators** (such as motors or hydraulic systems) to move or interact with objects.

Defining the Robot: Sensing, Thinking, Acting



VNR Lab Protocol

- It is through experiments that human learning has evolved by leaps and bounds in the history of science.
- Laboratories in an academic institute offer the perfect scenario for the young learners to get acquainted with the technological know-how and enable a hands-on experience of learning by doing.
- While traditional theory courses help the students with finding a foothold in the world of engineering, practical courses encourage the students to test and appreciate the theory.
- The long-term vision of a laboratory, in the today's Technological Age, shall inevitably be to create interest among students towards experimentation and research.
- Such a zeal can only be initiated through laboratory-based learning of theory, because it is in laboratories that students can realize the value of the theory and consider varying practical scenarios to test the concepts, laws, operation, and even hypotheses pertaining to the theory.
- An appreciably designed laboratory course must result in comprehensive and integrated learning and not simply teach numerous disconnected experiments.
- It is with this view that the VNR Lab Protocol had been designed for better understanding of laboratory experiments and their significance in practice of Engineering Profession.

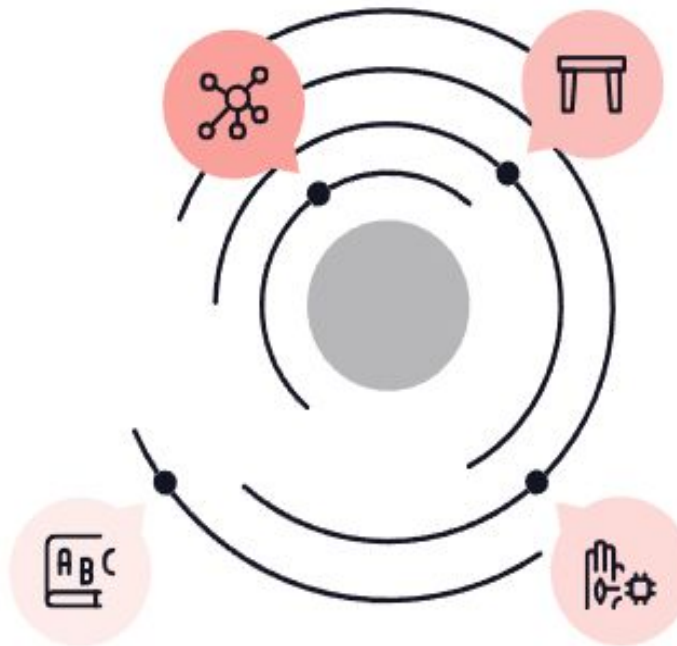
Objectives of VNR Lab Protocol

Interconnected Learning

View the lab as an "interconnected whole" where each experiment builds toward a larger robotic system.

Continuous Mapping

Every lab activity is mapped to your syllabus to enhance the depth of your theoretical understanding.



Design over Execution

You are encouraged to design and plan experiments rather than just following a manual step-by-step.

Manual Dexterity

Intellectual growth is paired with the physical skills required to handle sensitive electronic components.

COURSE

OBJECTIVES & OUTCOMES

:COURSE OBJECTIVES:

- Process Proprioceptive and Exteroceptive sensor data and analyze Sensor properties.
- Analyze the hardware differences between Rolling and Global shutter camera systems.
- Implement computer vision algorithms for object tracking and depth estimation.
- Control electric actuators (DC, Stepper, BLDC) and implement a closed loop perception-action system.

:COURSE OUTCOMES:

After completion of the course, the student should be able to

- **CO-1 (Analyze):** Characterize sensor noise & drift in IMUs and calibrate encoders for Odometry.
- **CO-2 (Apply):** Deploy vision hardware configurations to mitigate motion artifacts using Global and Rolling shutter technology.
- **CO-3 (Analyze):** Utilize LiDAR and Depth cameras to generate environmental maps & distance data.
- **CO-4 (Create):** Design integrated robotic system using vision-based feedback to control servo actuated mechanisms.

LIST OF EXPERIMENTS

Cycle I: Sensors & Perception

- **Experiment 1: Odometry and Kinematics:** Process quadrature encoder data from a live sensor or dataset to estimate the displacement and velocity of a robotic platform.
- **Experiment 2: Inertial Data Analysis:** Characterize noise and bias from an Inertial Measurement Unit (IMU) stream to estimate stable orientation via digital filtering.
- **Experiment 3: Ranging and Error Characterization:** Analyze Ultrasonic and Infrared distance data to model sensor accuracy and identify technology-specific limitations.
- **Experiment 4: Shutter and Motion Artifacts:** Evaluate image sequences from Rolling and Global shutter sources to quantify geometric distortion in dynamic scenes.
- **Experiment 5: Geometric Camera Calibration:** Utilize a dataset of calibration patterns to determine intrinsic parameters and rectify lens-induced distortions.
- **Experiment 6: Object Detection and Tracking:** Implement a vision pipeline to extract the spatial coordinates of a target from a live camera feed or a recorded video stream.

- **Experiment 7: Volumetric Depth Interpretation:** Process 3D point clouds or depth maps from a vision sensor to extract spatial geometry and distance metrics.
- **Experiment 8: Planar Environment Mapping:** Analyze 2D LiDAR scan data to generate polar plots and identify obstacles in a 2D plane.

Cycle II: Actuators & Control

- **Experiment 9: DC Motor Characterization:** Drive a DC motor through an H-Bridge while examining how its current draw responds to changing load levels.
- **Experiment 10: Stepper Motor Precision & Resonance:** Pair a stepper motor with a micro-stepping driver to assess the smoothness of its motion and any resulting vibration.
- **Experiment 11: Brushless DC (BLDC) Speed Control:** By linking a BLDC motor to an electronic controller, correlate the applied control signals with the resulting angular velocity.
- **Experiment 12: Integrated Visual Servoing:** Build a closed-loop arrangement in which visual feedback steers an actuated mechanism to follow a moving target.

STORY BOARD FOR SENSORS AND ACTUATORS LAB



Autonomous Mobile Robot (AMR) for Smart Warehouse Automation

Exp-1: Odometry and Kinematics

Wheel encoders estimate the robot's displacement, velocity, and travelled path within the warehouse.

Application: Determining how far the robot has moved from Rack A to Rack B.

Exp 2: Inertial Data Analysis

IMU data provides orientation and heading information.

Application: Maintaining stable navigation when the robot turns around corners or travels over uneven flooring.

Exp 3: Ranging and Error Characterization

Ultrasonic and IR sensors detect nearby obstacles and pallet positions.

Application: Preventing collisions with workers or warehouse shelves.

Exp 4: Shutter and Motion Artifacts

The onboard camera captures images while the robot is moving.

Application: Evaluating distortion effects during high-speed navigation and barcode scanning.

Exp 5: Geometric Camera Calibration

Accurate localization of objects and QR codes requires calibrated cameras.

Application: Identifying storage-bin locations with minimal measurement error.

Exp 6: Object Detection and Tracking

Detect packages, pallets, forklifts, and human workers.

Application: Tracking a moving pallet being transported by another robot.

Exp 7: Volumetric Depth Interpretation

Depth cameras generate 3D information about warehouse structures.

Application: Estimating package dimensions and detecting free storage space.

Exp 8: Planar Environment Mapping

LiDAR creates a 2D map of warehouse aisles.

Application: Real-time obstacle detection and route planning.

Exp 9: DC Motor Characterization

DC motors drive conveyor mechanisms and lifting systems.

Application: Analyzing power consumption while carrying different payloads.

Exp 10: Stepper Motor Precision and Resonance

Stepper motors position barcode scanners or robotic grippers.

Application: Accurate package alignment during sorting operations.

Exp 11: BLDC Speed Control

BLDC motors propel the mobile robot.

Application: Controlling speed while navigating narrow aisles.

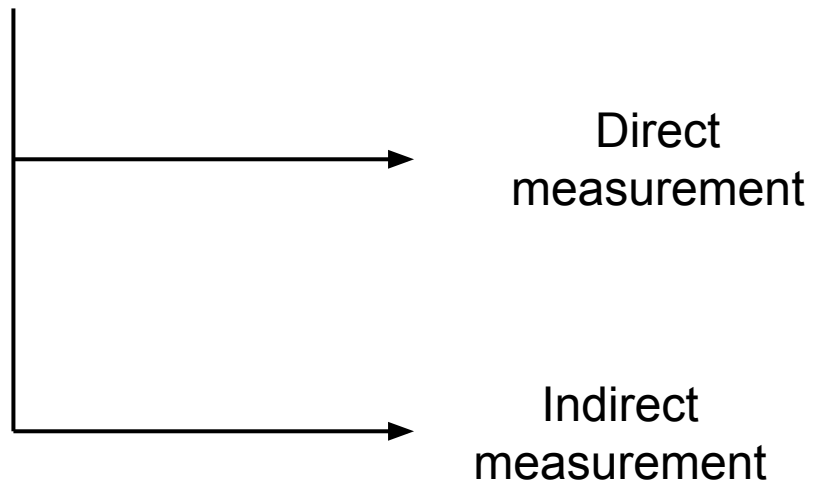
Exp 12: Integrated Visual Servoing

Camera feedback controls robot positioning.

Application: Automatically aligning the robot with a charging dock or storage rack.

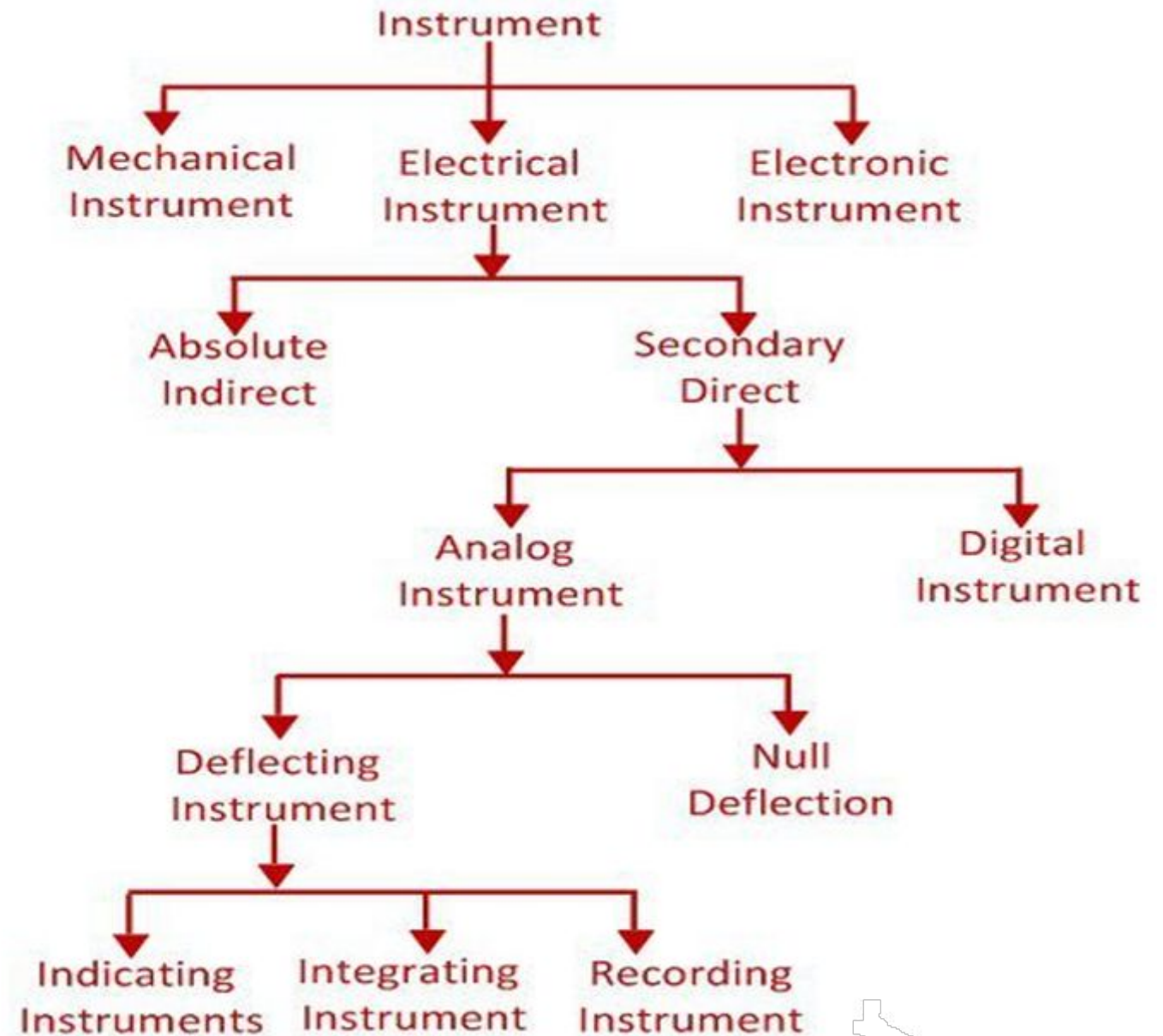
Basic Terminology

- **Instrumentation Engineering:** Branch of Engineering which deals with measurement and control of physical quantities.
- **Measurement:** Process of associating numbers with physical quantities/phenomena.



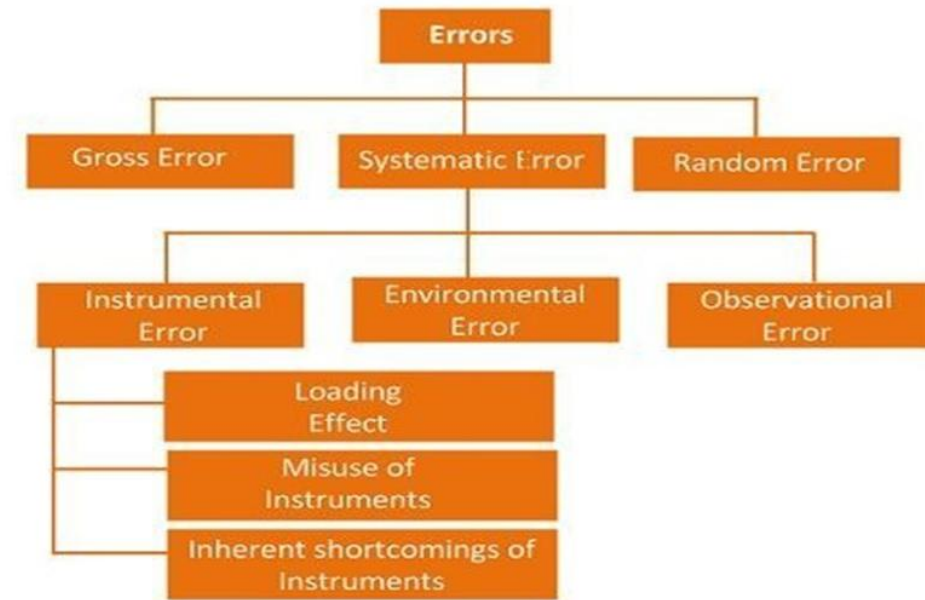
Direct vs Indirect Measurement		
Aspect	Direct Measurement	Indirect Measurement
Definition	Measurement obtained directly from an instrument	Measurement calculated from one or more direct
Method	Direct observation	Mathematical relationship or formula
Complexity	Simple	More complex
Calculation Required	No	Yes
Uncertainty Source	Instrument only	Multiple measured variables
Accuracy	Generally higher	Depends on calculation and input accuracy
Time Required	Faster	Slower
Typical Instruments	Vernier caliper, thermometer	Derived calculations using multiple instruments
Example	Measuring length with a ruler	Calculating density from mass and volume
ISO Context	Primary measurement	Derived measurement

- **Instrument:** Device used to perform measurement.
-

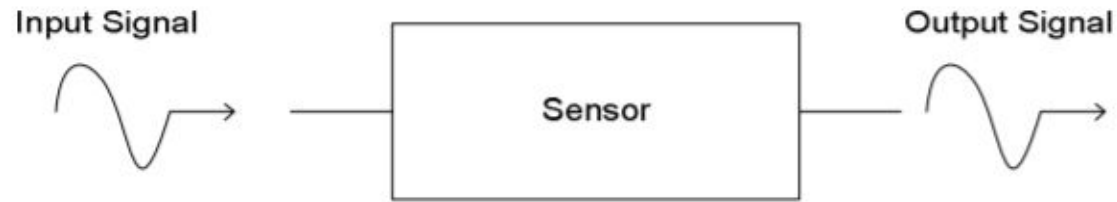


- **Absolute Instruments** are those which give the value of the quantity to be measured, in terms of the constants of the instrument and their deflection only.
- **Secondary Instruments** are those, in which the value of quantity to be measured can be determined from the deflection of the instruments, only when they have been pre-calibrated by comparison with an absolute instrument.
- An instrument in which zero or null indication determines the magnitude of measured quantity such type of instrument is called a **Null type** instrument.
- The instruments in which the measured quantity produces physical effects which deflect/displace the moving system of the instruments is known as **Deflection type** instrument.
- **Indicating instruments** indicate the quantity to be measured by means of a pointer which moves on a scale.
- **Recording instruments** record continuously the variation of quantity with respect to time. Here, a permanent continuous record of the indication is made on a chart/dial. Recording is generally made by a pen on a graph paper which is rotated on a disc or drum at a uniform speed. The amount of the quantity at any time (instant) may be read from the traced chart.
- **Integrating instruments** record the consumption of the total quantity of electricity, energy etc. during a particular period of time i.e. these instruments totalize events over a specified period of time. Examples are Ampere-hour meter, Kilo-watt-hour meter etc.

- **Error:** **Instrument error** refers to the difference between the actual value and the value indicated by the instrument
- **Gross Errors:** This category basically takes into account human oversight and other mistakes while reading, recording and readings.
- **Systematic Error:** It occurs due to identifiable sources.
 - 1) Observational errors occur when you make an incorrect observation.
 - 2) Instrumental error happen when an instrument gives wrong reading.
 - 3) Environmental errors are a result of the laboratory environment.
- **Random Errors:** Random errors are errors which occur irregularly and hence are random.

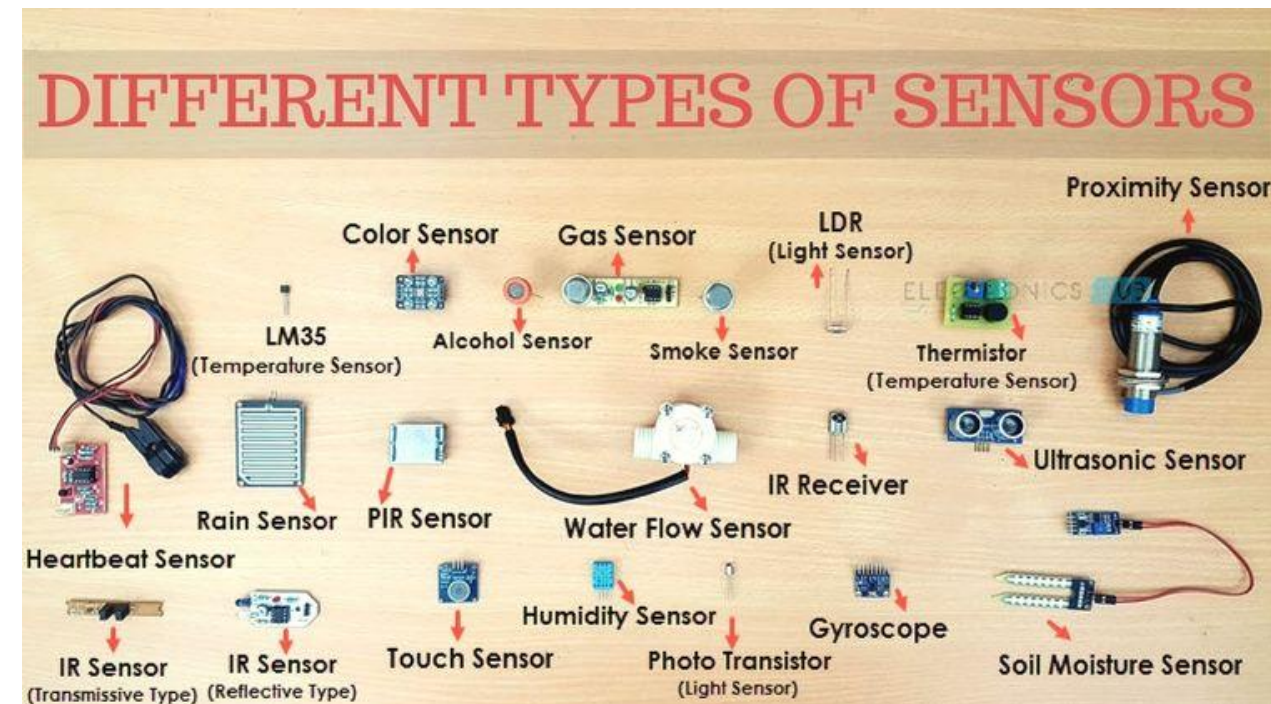


- **Sensor:** Device which is used to sense a physical quantity and produce suitable output



- **Advantages of Electronic sensors:**

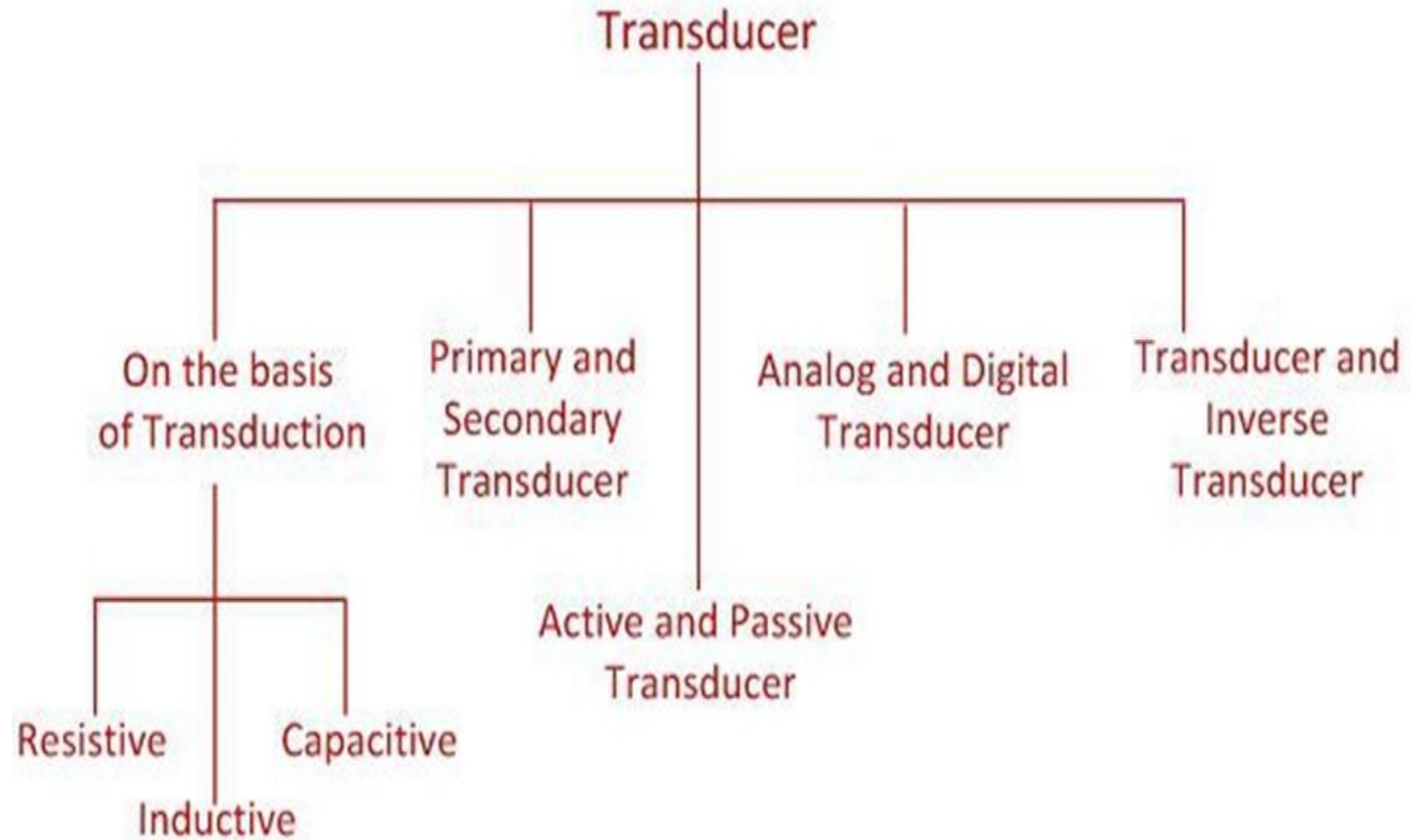
- 1) Amplification and attenuation can be done easily.
- 2) Mass inertia/Frictional effects are minimized.
- 3) Electronic systems can be controlled with very small electric power.
- 4) Electric power can be easily used, transmitted and processed.



- **Transducer:** A transducer is any device that converts one form of energy into a readable signal. Common inputs include energy, torque, light, force, position, acceleration etc.

Proprioceptive/Introceptive sensors: Measure values internal to the system (robot). Examples are motor speed, wheel load, robot arm joint angles, battery voltage.

Exteroceptive sensors: Acquire information from the robot's environment; Examples are distance measurements, light intensity, sound amplitude.



Factors to be considered while selecting transducer

- Transducer should have good resolution over the entire selected range.
- Transducer must be highly sensitive to desired signal and insensitive to unwanted signal.
- Transducer must be small in size/light in weight.
- Transducer should be able to work in corrosive environment.
- Transducer must withstand pressure, shocks, vibrations etc.
- Readings must be free from errors.
- The transducer circuit should have overload protection so that it will withstand overloads.

- **Actuators:** An actuator is a device which actuates or moves something. An actuator uses some type of energy to provide motion or to apply a force.

Types of Actuators

Electrical actuators

- Electric motors
 - DC servomotors
 - AC motors
 - Stepper motors
- Solenoids



Hydraulic actuators

- Use hydraulic fluid to amplify the control command signal



Pneumatic actuators

- Use compressed air as the driving force



Laboratory Guidelines

- Certain guidelines – most of which are common-sense – must be put in place in order to ensure a safe learning experience for young students. Above all, it is important for students to enjoy the laboratory experience and be prepared, observant, thinking, and anticipating during the course of the experiment.

- 1) Your total concentration on the experiment is required for a safe, meaningful laboratory experience.
- 2) Keep your face away from the apparatus. Maintain distance from circuits while they are under operation.
- 3) Do not power up a circuit until all connections are made and tested to be secure and approved by a faculty.
- 4) Do not lift any apparatus high into the air or away from the tables they are placed on. If you must do so, ensure that you disconnect all wires and cables from the apparatus and ensure that it does not have any wires and cables in its way to be lifted.

Laboratory Guidelines

- 5) Ensure that you have enough leg space and circulation area that is not blocked by anything such as book bags, stools & obsolete equipment. Keep your bags and belongings in the designated area.
- 6) Don't disturb your colleagues in the laboratory while they conduct experiments.
- 7) Everyone using the lab should know where the fire extinguisher is located and be familiar with the usage instructions.
- 8) Do not touch
 - (i) Exposed wires or sockets/open the trainer unit casing especially when it is turned ON
 - (ii) Spill water or any other fluids near the workspace
 - (iii) Tamper any of the hardware devices inappropriately.

Operational Conduct

- A sensible and professional attitude is expected of all students in the laboratory area. You may socialize with others in the lab, but do not go overboard. You are expected to maintain a learning, scientific environment.
- Listen to your laboratory instructor for additional guidelines, norms and rules.
- Use of mobile phones is prohibited in academic areas as per the instructions from the Institute's Administration. Please understand that lab also come under academic areas.
- Clear your laboratory workspace when you leave, and arrange the chairs properly.
- You are most welcome to come work in the laboratory when you have free time and when there is no ongoing laboratory session. However, please ensure that a faculty member/instructor knows what you are doing and that someone of them is around to supervise or guide you all through your stay in the laboratory

Experimental Conduct

- 1) Review the Objectives and Introduction of the experiment. A thorough knowledge of the experiment will make the laboratory experience more time efficient and scientifically more meaningful.
- 2) Review the Experimental Procedure. Try to understand the purpose of each step.
- 3) Determine if any extra equipment is needed and be ready to obtain it all at once from the instructor.
- 4) Determine the data to be collected and how to analyze/present it.
- 5) Scientists learn much by discussion with one another. Likewise, you may profit by discussion with your laboratory instructor or classmates.
- 6) Switch off all individual equipment and then switch off the power supply to circuit(s) at the end of each laboratory session.

Data – Documentation, Graphing, Analysis

- Science can form conclusive discoveries and inventions and formulate fundamental laws only on the basis of reliable and reproducible data.
- Data thus lay the foundation for all learning in science and technology.
- It is important for all experimenters to identify significant parameters in each experiment and make a note of all such data in such a way that subsequent analyses lead to believable conclusions, which can be presented through graphs or plots leading to a better understanding of the experimental results.
- Confidence in a scientific theory depends on the reliability of the experimental data on which the theory is based.
- In order to obtain reliable data, all instruments must be calibrated appropriately for a properly designed experimental procedure.

- Systematic errors must be identified and addressed appropriately for good accuracy and precision.
- Human errors in collecting and recording data must be avoided too, while also taking proper care to ensure that all apparatus are used in the recommended ways only.
- Random errors can also occur in measurements, and repeated readings shall help ensure that they do not affect the reliability of the data considerably.
- A well-designed graph of experimental data is a very effective organization of the data for observing trends, discovering relationships, or predicting information.
- It is therefore worthwhile to learn how to effectively construct and present a graph and how to extract information from it.

Documentation

- 1) When starting an experiment, clearly outline the following information:
 - a. Title of Experiment/Date of Conduction
 - b. Aim/Objective/Purpose of the Experiment
 - d. Experimental Procedure including circuit diagrams/Specific precautions and Safety concerns.
- 2) Your laboratory record of observations is your personal, permanent record of the activities associated with the experiment and other laboratory activity.
- 3) Record all parameters of interest, starting from the input end and proceeding to the output.
- 4) Write down the appropriate units for all the fixed as well as varying parameters.
- 5) Record the data with the precision shown on the appropriate display.

Graph and Analysis

- 1) Calculations must be carried out in prescribed space and must be shown in detailed steps as necessary.
- 2) Calculations must indicate units also for all parameters.
- 3) When graphing the data:
 - a) Choose the appropriate axes – independent variables may be plotted on the abscissa while dependent variable is plotted on the ordinate.
 - b) Label the axes appropriately, indicating the units/Choose a reasonable scale for each axis.
 - c) Indicate the scale of the axes on the top right corner of the plot area.
 - d) Place a dot for each data point and draw a small circle around it. (Ideally, the size of the circle indicates the estimated error in the measurement).
 - e) Draw a smooth curve such that it passes through closely through the whole dataset.
 - f) Identify every plot by an appropriate title indicating what the plot represents.

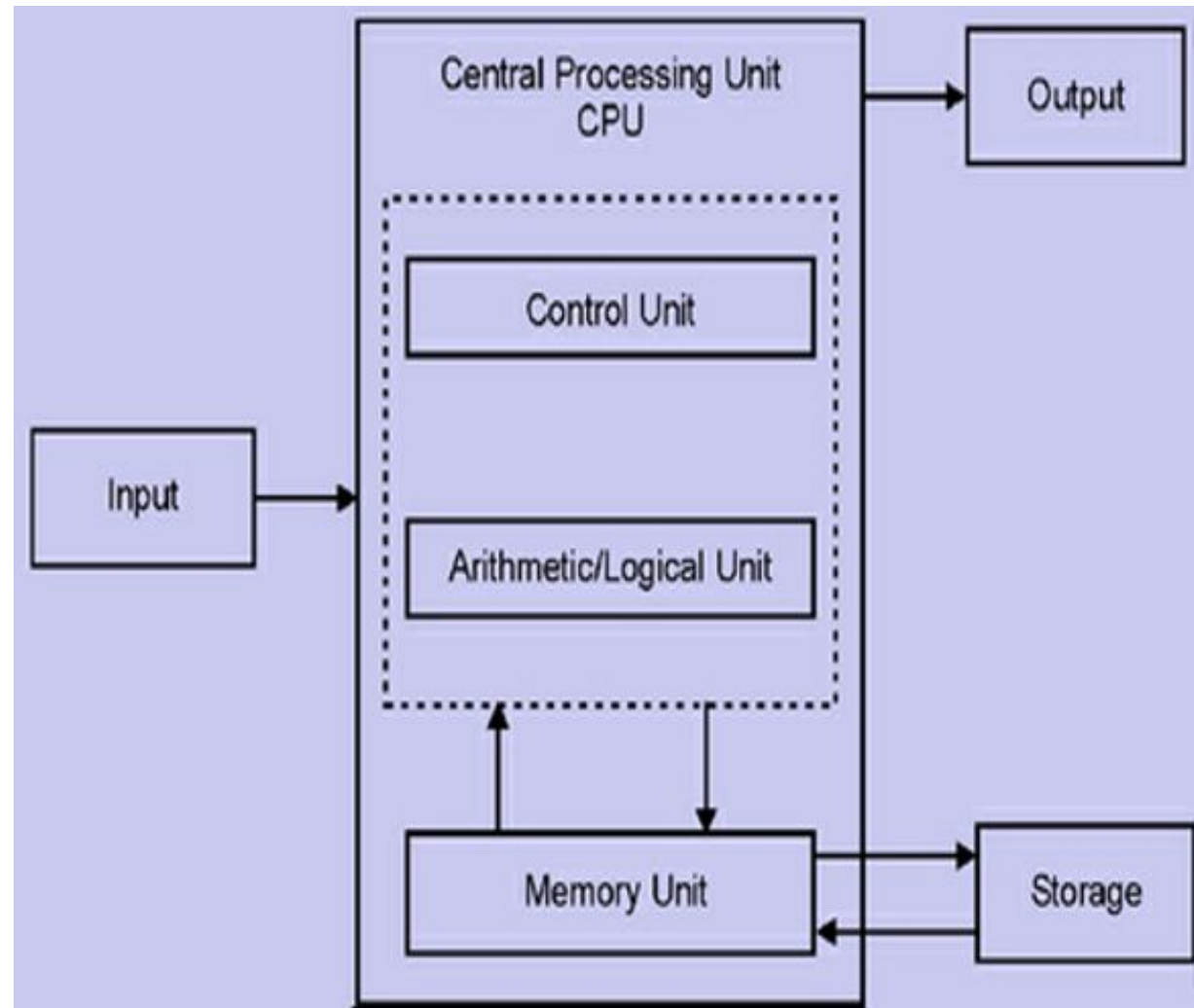
Introduction To Arduino & Raspberry Pi

- Robotics is an interdisciplinary field combining engineering, computer science and technology to design, construct and operate machines that replicate or assist human actions.
- Modern robotics integrates mechanical design, electronics and advanced software(heavily relying on artificial intelligence) to enable machines to perceive, think and act autonomously.

Key Concepts

- **Sensing:** Robots gather data about their surroundings through specialized hardware like cameras, LiDAR and sensors.
- **Thinking:** Using microcontrollers, processors and AI algorithms, the robot interprets sensory data and decides on the appropriate action.
- **Acting:** The machine physically carries out the task using motors, hydraulic systems and actuators.

Basic Computing Diagram



- **Arduino** is a microcontroller ideal for reading sensors and controlling hardware in real-time.
- **Raspberry Pi** is a fully functional computer designed to process data, run software, and connect to the internet. Using them together combines reliable, low-level hardware control with high-level computing and networking.

When to Use Arduino

Arduino boards excel at tasks requiring direct, immediate interaction with electronic components.

- **Real-time processing:** It executes simple commands deterministically, without an operating system slowing it down.
- **Hardware control:** It natively reads analog sensors (e.g., temperature, light) and directly powers motors or LEDs using dedicated pins.
- **Low power:** They draw very little current and can easily run for months on a battery.

When to Use Raspberry Pi

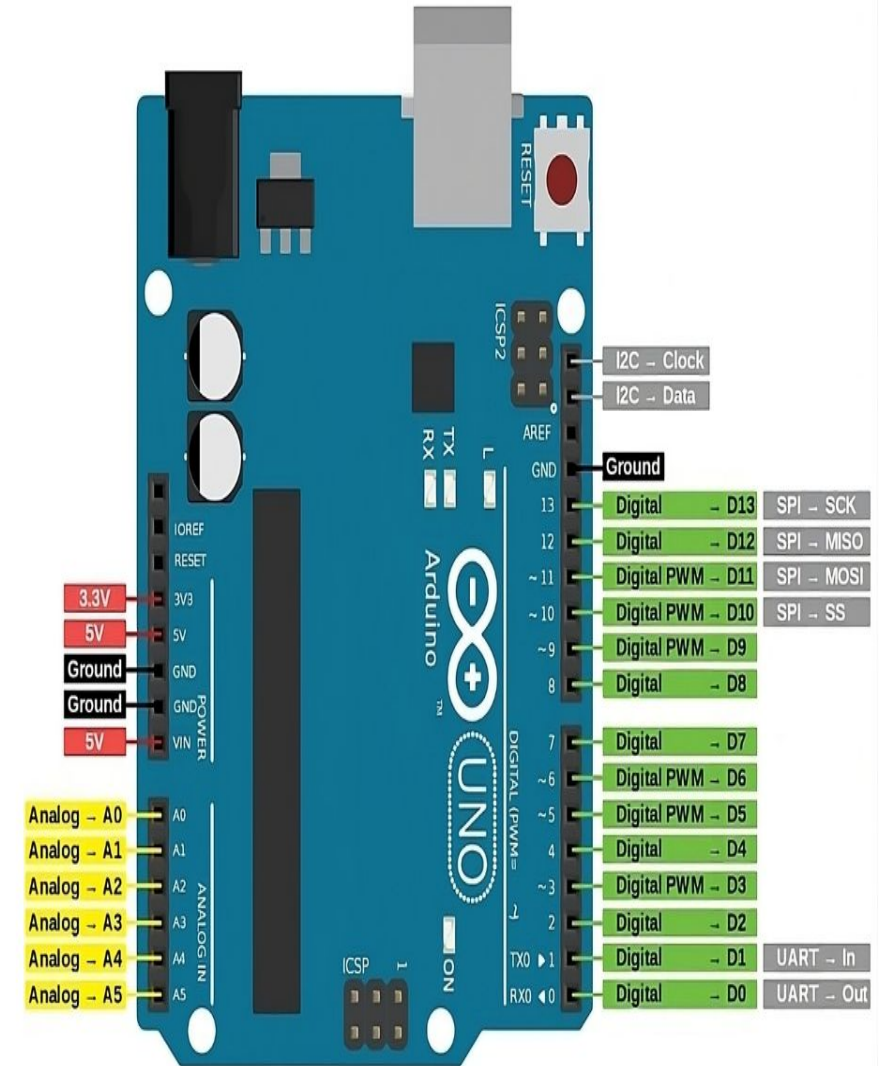
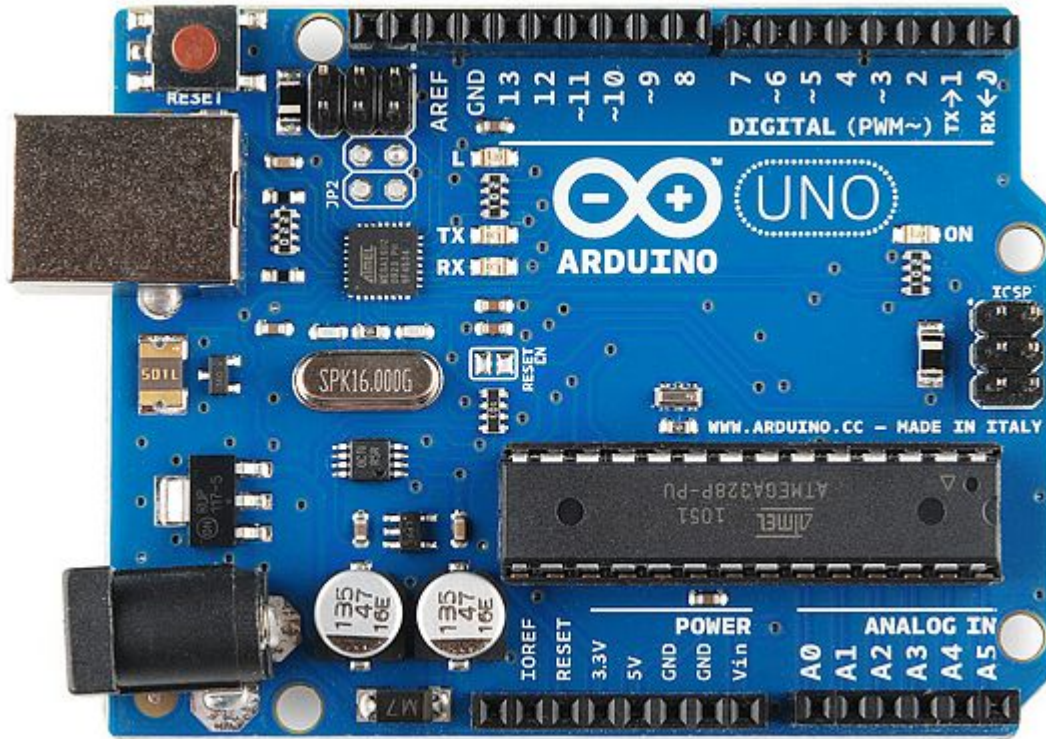
Raspberry Pi acts as a mini-computer, featuring multiple USB ports, built-in Wi-Fi, and the ability to run full Linux-based operating systems.

- **High-level computing:** It has the processing power to handle tasks like machine learning, computer vision, and facial recognition.
- **Networking and Servers:** Ideal for hosting a personal website, media center, or acting as a game server.
- **User interfaces:** Easily connects to a monitor, keyboard, and mouse to provide a desktop-like experience.

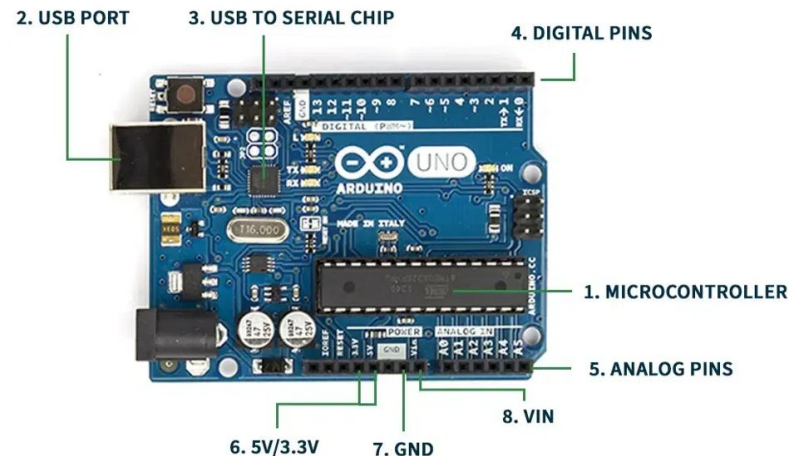
Feature	Arduino Uno	Raspberry Pi
Type	Microcontroller (ATmega328P)	Single-board Computer (Linux OS)
Best For	Real-time hardware control	Computer Vision & AI
Sensors	Native Analog reading	Digital via GPIO / USB
Power	Ultra-low, battery efficient	Higher consumption
Language	C++ (Arduino IDE)	Python, C++, Java

ARDUINO

- An Arduino is an open-source electronics platform used to build interactive projects, prototypes, and devices. It acts as a programmable "brain" that reads inputs (like pressing a button or reading a temperature) and controls outputs (like turning on a motor, an LED, or a screen).



- **Microcontroller:** Controls all major operations. The microcontroller is used to coordinate the input taken and execute the code written in a high-level language.
- **Analog Reference pin:** Used for general purposes like supporting 10-bit analog-to-digital conversion (ADC) which is performed using analog the Read() function. Analog pins are particularly helpful since they can store 0-255 bits which is not possible using digital pins.
- **Digital Pins:** Used for general purposes like taking input or generating output. The commands that are used for setting the modes of the pins are pinMode(), digitalRead(), and digitalWrite() commands.
- **Reset Button:** Used for setting all the components of Arduino to their default values. In case you want to stop the Arduino in between you can use this reset button.
- **Power and Ground Pins:** Supply the power needed for driving the Arduino board. The ground pins are usually 0V to set a reference level for the circuit.
- **USB (universal serial bus):** The Arduino needs certain protocols for communication purposes and the universal serial bus is used for this purpose. It helps to connect Arduino, microcontrollers with other raspberry pies.



Essential items needed to work with ARDUINO:

- **ARDUINO Board:** The ARDUINO UNO Board.
- **Breadboard & Jumper Wires:** For connecting electronic components.
- **Sensors & Actuators:** LEDs, buzzers, sensors and small motors.
- **USB Cable:** To connect the board to PC so as to upload the code.

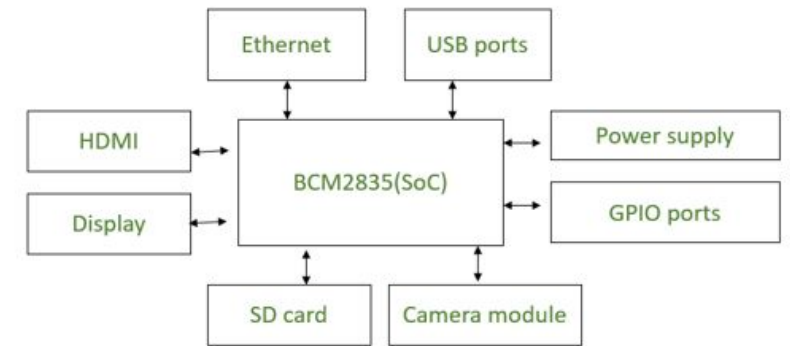
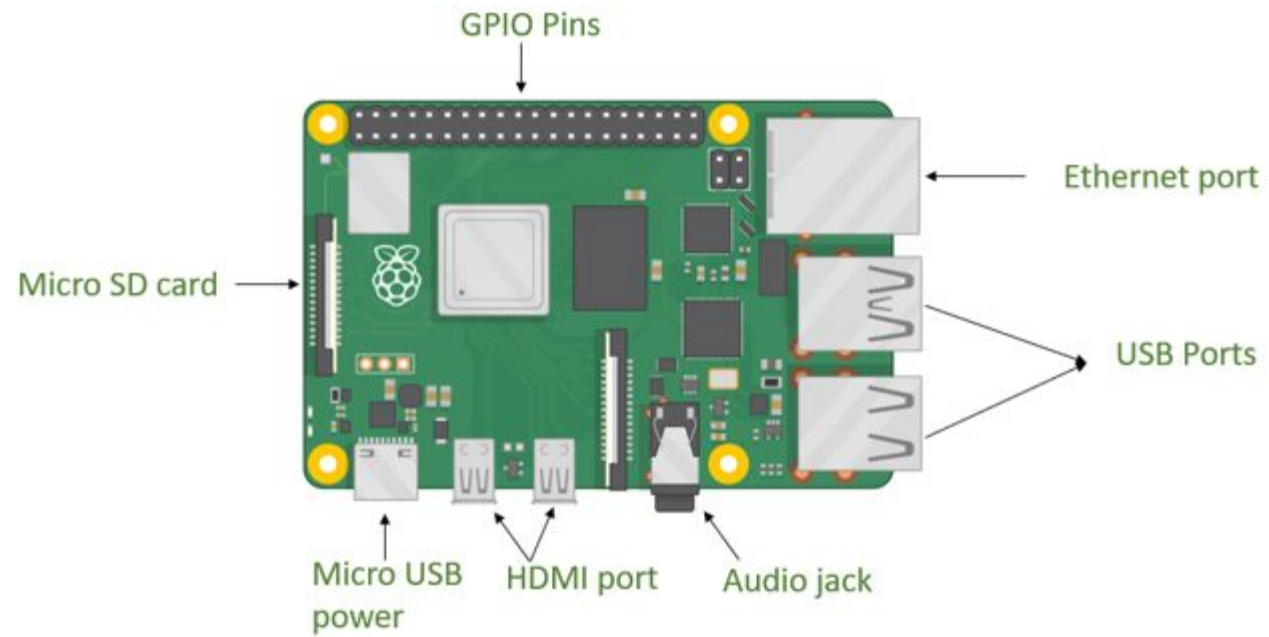
Overview of Arduino UNO

- Open-source microcontroller board based on ATmega328P
- 14 digital I/O pins, 6 analog inputs
- 16 MHz clock, USB programming
- Ideal for real-time embedded control
- Easy programming using Arduino IDE
- Low cost and low power
- **Fast response, limited memory.**

Raspberry Pi

- Small size computer.
- Runs on several operating systems.
- Possible to connect with a variety of sensors to interact with the physical world
- Make decisions based on processing gathered sensor data





Architecture of Raspberry Pi

Raspberry Pi mainly consists of the following blocks:

- **Processor:** Raspberry Pi uses Broadcom BCM2835 system on chip which is an ARM processor and Video core Graphics Processing Unit (GPU). It is the heart of the Raspberry Pi which controls the operations of connected devices and handles all the required computations.
- **HDMI:** High Definition Multimedia Interface is used for transmitting video or digital audio data to a computer monitor or to digital TV. This HDMI port helps Raspberry Pi to connect its signals to any digital device such as a monitor digital TV or display through an HDMI cable.
- **GPIO ports:** General Purpose Input Output ports are available on Raspberry Pi which allows the user to interface various I/P devices.
- **Audio output:** Audio connector connects audio output devices such as headphones/peakers.
- **USB ports:** This is a common port available for various peripherals such as a mouse, keyboard, or any other I/P device. With the help of a USB port, the system can be expanded by connecting more peripherals.

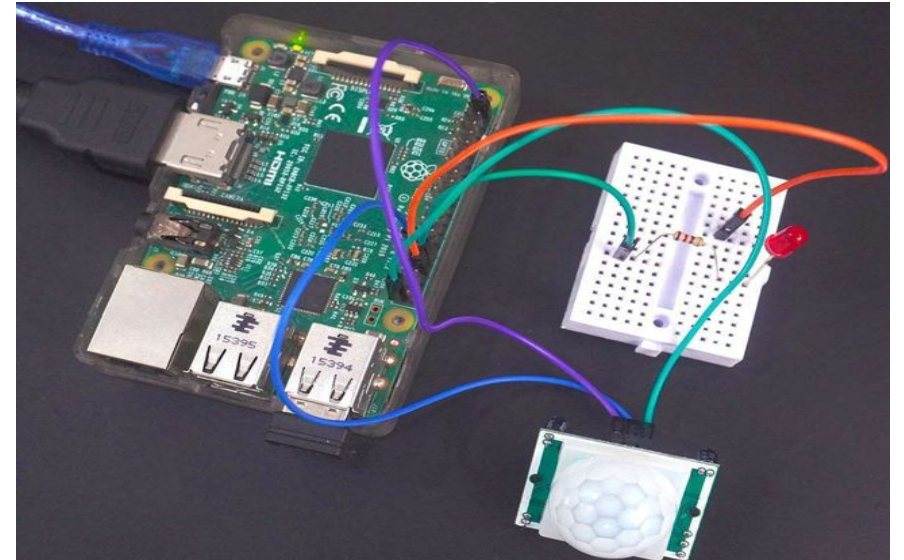
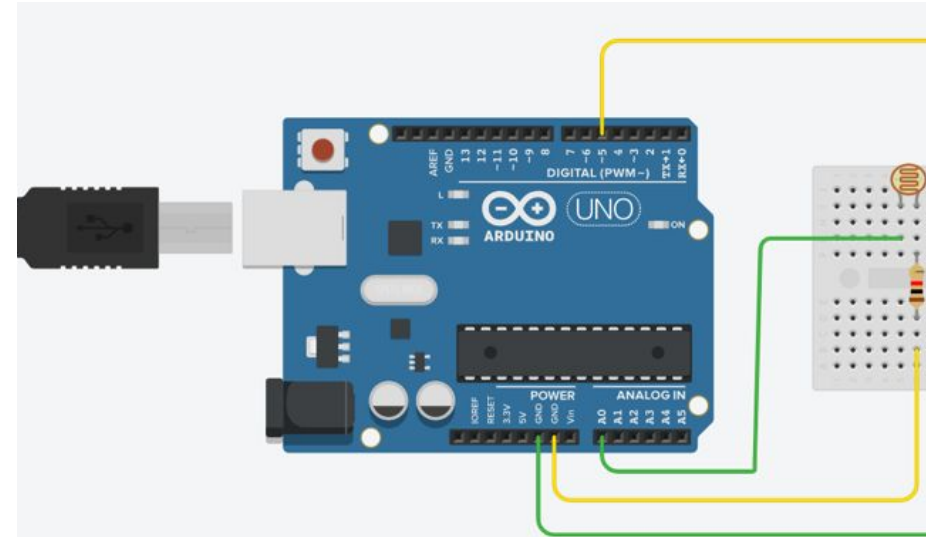
- **SD card:** The SD card slot is available on Raspberry Pi. An SD card with an operating system installed is required for booting the device.
- **Ethernet:** The ethernet connector allows access to the wired network, it is available only on the model B of Raspberry Pi.
- **Power supply:** A micro USB power connector is available onto which a 5V power supply can be connected.
- **Camera module:** Camera Serial Interface (CSI) connects the Broadcom processor to the Pi camera.
- **Display:** Display Serial Interface (DSI) is used for connecting LCD to Raspberry Pi using 15 15-pin ribbon cables. DSI provides a high-resolution display interface that is specifically used for sending video data.

Overview of Raspberry Pi

- Single-board computer/ARM-based processor
- Runs on Linux OS
- HDMI, USB, Wi-Fi/Bluetooth enabled
- Multi-core processor
- GPIO interface
- Python, C/C++, Java support
- Networking and cloud connectivity

- **Powerful computing and ideal for simple real-time tasks**

Interfacing Diagram



Continuous Internal Evaluation (CIE)

1) Each Practical course shall carry **FORTY (40) marks** for CIE, distributed as:

10 Marks

Day-to-Day Perform.

Active participation and preparation for each lab session.

10 Marks

Record Work

Accurate documentation, graphing, and analysis of experimental data.

10 Marks

Practical Exam

Demonstrating technical proficiency in a controlled setting.

10 Marks

Course Based Project

Developing a hardware prototype or simulation that integrates lab concepts.

2) The course project component may include computational modeling, hardware implementation, design, prototype development, simulation, software/application development, or any experiential learning activity.

3) Absence in any CIE component shall result in zero marks for that component.

4) No reassessment shall be permitted.



THANK YOU
FOR
YOUR
ATTENTION
ANY QUESTIONS?